which corresponds to the exponential weights algorithm on the estimated losses $\widetilde{\boldsymbol{\ell}}^s$. Apply (1.41) to the estimated losses to show that for any $\pi \in [A]$,

$$\mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t} \langle p^t, \widetilde{\boldsymbol{\ell}}^t \rangle\right] - \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t} \langle e_\pi, \widetilde{\boldsymbol{\ell}}^t \rangle\right] \lesssim \sqrt{AT \log A}$$

Hence, the price of bandit feedback in the adversarial model, as compared to full-information online learning, is only $\sqrt{A}$.

3. CONTEXTUAL BANDITS

In the last section, we studied the multi-armed bandit problem, which arguably the simplest framework for interactive decision making. This simplicity comes at a cost: few real-world problems can be modeled as a multi-armed bandit problem directly. For example, for the problem of selecting medical treatments, the multi-armed bandit formulation presupposes that one treatment rule (action/decision) is good for all patients, which is clearly unreasonable. To address this, we augment the problem formulation by allowing the decision-maker to select the action π^t after observing a *context* x^t; this is called the $\boxed{\textit{contextual bandit} \text{ prob-}}$ lem. The context x^t, which may also be thought of as a feature vector or collection of

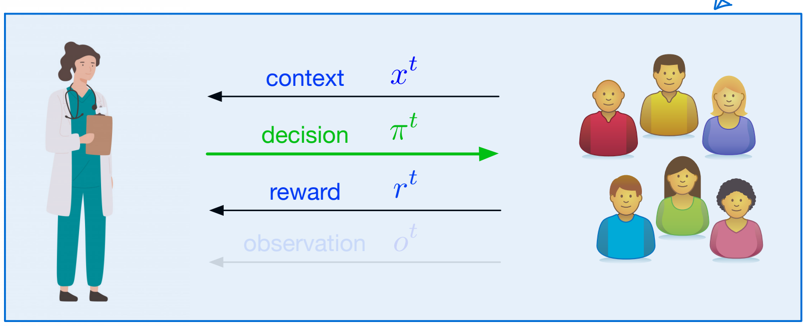


Figure 6: An illustration of the contextual multi-armed bandit problem. A doctor (the learner) aims to select a treatment based on the context (medical history, symptoms).

covariates (e.g., a patient's medical history, or the profile of a user arriving at a website), can be used by the learner to better maximize rewards by tailoring decisions to the specific patient or user under consideration.

Contextual Bandit Protocol
for $t = 1, \ldots, T$ **do**
  Observe context $x^t \in \mathcal{X}$.
  Select decision $\pi^t \in \Pi = \{1, \ldots, A\}$.   Policy .
  Observe reward $r^t \in \mathbb{R}$.

As with multi-armed bandits, contextual bandits can be studied in a stochastic framework or in an adversarial framework. In this course, we will allow the contexts $x^1, \ldots, x^T$ to be generated in an arbitrary, potentially adversarially fashion, but assume that rewards are generated from a fixed conditional distribution.

Assumption 2 (Stochastic Rewards): Rewards are generated independently via

$$r^t \sim M^\star(\cdot \mid x^t, \pi^t), \tag{3.1}$$

where $M^\star(\cdot \mid \cdot, \cdot)$ is the underlying *model* (or conditional distribution).

This generalizes the stochastic multi-armed bandit framework in Section 2. We define

$$f^\star(x, \pi) := \mathbb{E}\left[r \mid x, \pi\right] \tag{3.2}$$

as the mean reward function under $r \sim M^\star(\cdot \mid x, \pi)$, and define $\pi^\star(x) := \arg\max_{\pi \in \Pi} f^\star(x, \pi)$ as the optimal *policy*, which maps each context x to the optimal action for the context. We measure performance via regret relative to $\pi^\star$:

$$\mathbf{Reg} := \sum_{t=1}^{T} f^\star(x^t, \pi^\star(x^t)) - \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(x^t, \pi^t)], \tag{3.3}$$

where $p^t \in \Delta(\Pi)$ is the learner's action distribution at step t (conditioned on the $\mathcal{H}^{t-1}$ and x^t). This provides a (potentially) much stronger notion of performance than what we considered for the multi-armed bandit: Rather than competing with the reward of the single best action, we are competing with the reward of the best sequence of decisions tailored to the context sequence we observe.

Remark 12 (Contextual bandits versus reinforcement learning): To readers already familiar with reinforcement learning, the contextual bandit setting may appear quite similar at first glance, with the term "context" replacing "state". 비슷 The key difference is that in reinforcement learning, we aim to control the evolution of $x^1, \ldots, x^T$ (which is why they are referred to as state), whereas in contextual bandits, we take the sequence as a given, and only aim to maximize our rewards conditioned on the sequence.

원하는 것, 필요한 것

Function approximation and desiderata. If $\mathcal{X}$, the set of possible contexts, is finite, one might imagine running a separate MAB algorithm for each context. In this case, the regret bound would scale with $|\mathcal{X}|$,[9] an undesirable property which reflects the fact that this approach does not allow for generalization across contexts. Instead, we would like to share information between different contexts. After all, a doctor prescribing treatments might never observe exactly the same medical history and symptoms twice, but they might see similar patients or recognize underlying patterns. In the spirit of statistical learning (Section 1) this means assuming access to a class $\mathcal{F}$ that can model the mean reward function, and aiming for regret bounds that scale with $\log|\mathcal{F}|$ (reflecting the statistical capacity of $\mathcal{F}$), with *no dependence* on the cardinality of $\mathcal{X}$. To facilitate this, we will assume a well-specified/realizable model.

[9] One can show that running an independent instance of UCB for each context leads to regret $\widetilde{O}(\sqrt{AT \cdot |\mathcal{X}|})$; see Exercise 5.

context x decision.

Assumption 3: The decision-maker has access to a class $\mathcal{F} \subset \{f : \boxed{\mathcal{X} \times \Pi \to \mathbb{R}}\}$ such that $f^\star \in \mathcal{F}$.

Using the class $\mathcal{F}$, we would like to develop algorithms that can model the underlying reward function for better decision making performance. With this goal in mind, it is reasonable to try leveraging the algorithms and respective guarantees we have already seen for statistical and online supervised learning. At this point, however, the decision-making problem—with its exploration-exploitation dilemma—appears to be quite distinct from these supervised learning frameworks. Indeed, naively applying supervised learning methods, which do not account for the interactive nature of the problem, can lead to failure, as we saw with the greedy algorithm in Section 2.1. In spite of these apparent difficulties, in the next few lectures, we will show that it is possible to leverage supervised learning methods to develop provable decision making methods, thereby bridging the two methodologies.

① e-e dilemma.
② greedy Algorithm.

① optimism

3.1 Optimism: Generic Template

What algorithmic principles should we employ to solve the contextual bandit problem? One approach is to adapt solutions from the multi-armed bandit setting. There, we saw that the principle of *optimism* (in particular, the UCB algorithm) led to (nearly) optimal rates for bandits, so a natural question is whether optimism can be adapted to give optimal guarantees in the presence of contexts. The answer to this last question is: *it depends*. We will first describe some positive results under assumptions on $\mathcal{F}$, then provide a negative example, and finally turn to an entirely different algorithmic principle.

Optimism via confidence sets. Let us describe a general approach (or, template) for applying the principle of optimism to contextual bandits [26, 1, 74, 37]. Suppose that at each time, we have a way to construct a *confidence set*

$$\mathcal{F}^t \subseteq \mathcal{F}$$

based on the data observed so far, with the important property that $f^\star \in \mathcal{F}^t$. Given such a confidence set we can define upper and lower *confidence functions* $\underline{f}^t, \bar{f}^t : \mathcal{X} \times \Pi \to \mathbb{R}$ via

$$\underline{f}^t(x,\pi) = \min_{f \in \mathcal{F}^t} f(x,\pi), \quad \bar{f}^t(x,\pi) = \max_{f \in \mathcal{F}^t} f(x,\pi). \tag{3.4}$$

These functions generalize the upper and lower confidence bounds we constructed in Section 2. Since $f^\star \in \mathcal{F}^t$, they have the property that

$$\underline{f}^t(x,\pi) \le f^\star(x,\pi) \le \bar{f}^t(x,\pi) \tag{3.5}$$

Property that $f^\star \in \mathcal{F}^t$

for all $x \in \mathcal{X}, \pi \in \Pi$. As such, if we consider a contextual analogue of the UCB algorithm, given by

$$\pi^t = \arg\max_{\pi \in \Pi} \bar{f}^t(x^t,\pi), \tag{3.6}$$

then as in Lemma 7, the optimistic action satisfies

x를 step t로 고정.

$$f^\star(x^t,\pi^\star) - f^\star(x^t,\pi^t) \le \bar{f}^t(x^t,\pi^t) - \underline{f}^t(x^t,\pi^t).$$

$$\mathbf{Reg} := \sum_{t=1}^{T} f^\star(x^t, \pi^\star(x^t)) - \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(x^t, \pi^t)],$$
$$= \sum_{t=1}^{T} \left(f^\star(x^t, \pi^\star) - f^\star(x^t, \pi^t) \right)$$ deterministic
$$\leq \sum_{t=1}^{T} \bar{f}^t(x^t, \pi^t) - \underline{f}^t(x^t, \pi^t)$$

That is, the suboptimality is bounded by the width of the confidence interval at (x^t, π^t), and the total regret is bounded as

$$\boxed{\mathbf{Reg} \leq \sum_{t=1}^{T} \bar{f}^t(x^t, \pi^t) - \underline{f}^t(x^t, \pi^t).} \tag{3.7}$$

To make this approach concrete and derive sublinear bounds on the regret, we need a way to construct the confidence set $\mathcal{F}^t$, ideally so that the width in (3.7) shrinks as fast as possible.

Constructing confidence sets with least squares. We construct confidence sets by appealing to a supervised learning method, empirical risk minimization with the square loss (or, least squares). Assume that $f(x, a) \in [0, 1]$ for all $f \in \mathcal{F}$, and that $r^t \in [0, 1]$ almost surely. Let

ERM.

$$\widehat{f}^t = \arg\min_{f \in \mathcal{F}} \sum_{i=1}^{t-1} (f(x^i, \pi^i) - r^i)^2 \tag{3.8}$$

to be large enough.

be the empirical risk minimizer at round t, and with $\beta := 8\log(|\mathcal{F}|/\delta)$ define $\mathcal{F}^1 = \mathcal{F}$ and

$$\mathcal{F}^t = \left\{ f \in \mathcal{F} : \sum_{i=1}^{t-1} (f(x^i, \pi^i) - r^i)^2 \leq \sum_{i=1}^{t-1} (\widehat{f}^t(x^i, \pi^i) - r^i)^2 + \beta \right\} \tag{3.9}$$

for $t > 1$. That is, our confidence set $\mathcal{F}^t$ is the collection of all functions that have empirical squared error close to that of $\widehat{f}^t$. The idea behind this construction is to set β "just large enough", to ensure that we do not accidentally exclude $f^\star$, with the precise value for β informed by the concentration inequalities we explored in Section 1. The only catch here is that we need to use variants of these inequalities that handle dependent data, since x^t and π^t are not i.i.d. in (3.8). The following result shows that $\mathcal{F}^t$ is indeed valid and, moreover, that all functions $f \in \mathcal{F}^t$ have low estimation error on the history.

Lemma 10: Let $\pi^1, \ldots, \pi^T$ be chosen by an arbitrary (and possibly randomized) decision-making algorithm. With probability at least $1 - \delta$, $f^\star \in \mathcal{F}^t$ for all $t \in [T]$. Moreover, with probability at least $1 - \delta$, for all $\tau \leq T$, all $f \in \mathcal{F}^\tau$ satisfy

mismatch.

$$\sum_{t=1}^{\boxed{\tau-1}} \mathbb{E}_{\pi^t \sim p^t}\left[(f(x^t, \pi^t) - f^\star(x^t, \pi^t))^2 \right] \leq 4\beta, \tag{3.10}$$

where $\beta = 8\log(|\mathcal{F}|/\delta)$.

Lemma 10 is valid for any algorithm, but it is particularly useful for UCB as it establishes the validity of the confidence bounds as per (3.5); however, it is not yet enough to show that the algorithm attains low regret. Indeed, to bound the regret, we need to control the confidence widths in (3.7), but there is a mismatch: for step τ, the regret bound in (3.7) considers the width at (x^τ, π^τ), but (3.10) only ensures closeness of functions in $\mathcal{F}^\tau$ under $(x^1, \pi^1), \ldots, (x^{\tau-1}, \pi^{\tau-1})$. We will show in the sequel that for linear models, it is possible to control this mismatch, but that this is not possible in general.

(3.11)

$(f-r)^2-(f^*-r)^2 = f^2-f^{*2}-2r(f-f^*)$

take $\mathbb{E}_{t-1}$, using $\mathbb{E}_{t-1}(r^t)=f^*$ RL property.

then. $\mathbb{E}_{t-1}(f^2-f^{*2}-2f^*(f-f^*)) = \mathbb{E}_{t-1}(f^2+f^{*2}-2f^*f)$

$= \mathbb{E}_{t-1}(f-f^*)^2$

(3.12)

Proof of Lemma 10. For $f \in \mathcal{F}$, define

$$U^t(f) = (f(x^t, \pi^t) - r^t)^2 - (f^\star(x^t, \pi^t) - r^t)^2. \tag{3.11}$$

It is straightforward to check that[10]

$$\mathbb{E}_{t-1} U^t(f) = \mathbb{E}_{t-1}(f(x^t, \pi^t) - f^\star(x^t, \pi^t))^2, \tag{3.12}$$

where $\mathbb{E}_{t-1}[\cdot] := \mathbb{E}[\cdot \mid \mathcal{H}^{t-1}, x^t]$. Then $Z^t(f) = \mathbb{E}_{t-1} U^t(f) - U^t(f)$ is a martingale difference sequence and $\sum_{t=1}^{\tau} Z^t(f)$ is a martingale. Since increments $Z^t(f)$ are bounded as $|Z^t(f)| \le 1$ (this holds whenever $f \in [0,1], r^t \in [0,1]$), according to Lemma 35 with $\eta = \frac{1}{8}$, with probability at least $1-\delta$, for all $\tau \le T$,

Lemma 35 (Freedman's inequality (Bernstein for martingales)): Let $(X_t)_{t \le T}$ be a real-valued martingale difference sequence adapted to a filtration $(\mathscr{F}_t)_{t \le T}$. If $|X_t| \le R$ almost surely, then for any $\eta \in (0, 1/R)$, with probability at least $1-\delta$, for all $T' \le T$,

$$\sum_{t=1}^{T'} X_t \le \eta \sum_{t=1}^{T'} \mathbb{E}_{t-1}[X_t^2] + \frac{\log(\delta^{-1})}{\eta}.$$

$$\sum_{t=1}^{\tau} Z^t(f) \le \frac{1}{8} \sum_{t=1}^{\tau} \mathbb{E}_{t-1}\big[Z^t(f)^2\big] + 8\log(\delta^{-1}). \tag{3.13}$$

$(f-r)^2-(f^*-r)^2 = (f-f^*)(f+f^*-2r)$

here. $f+f^*-2r \le 2$.

To control the right-hand side, we again use that $f, r^t \in [0,1]$ to bound

$$\mathbb{E}_{t-1}\big[Z^t(f)^2\big] \le \mathbb{E}_{t-1}\Big[\big((f(x^t,\pi^t) - r^t)^2 - (f^\star(x^t,\pi^t) - r^t)^2\big)^2\Big] \tag{3.14}$$

$$\le 4\,\mathbb{E}_{t-1}\big[(f(x^t,\pi^t) - f^\star(x^t,\pi^t))^2\big] = 4\,\mathbb{E}_{t-1} U^t(f) \tag{3.15}$$

Then, after rearranging, (3.13) becomes

$$0 \le \boxed{\frac{1}{2} \sum_{t=1}^{\tau} \mathbb{E}_{t-1} U^t(f)} \le \sum_{t=1}^{\tau} \underbrace{U^t(f)}_{(f-r)^2-(f^*-r)^2} + 8\log(\delta^{-1}). \tag{3.16}$$

$Z \to U$

Since the left-hand side is nonnegative, we conclude that with probability at least $1-\delta$,

$$\sum_{t=1}^{\tau} (f^\star(x^t,\pi^t) - r^t)^2 \le \sum_{t=1}^{\tau} (f(x^t,\pi^t) - r^t)^2 + 8\log(\delta^{-1}). \tag{3.17}$$

take $\delta \to \frac{\delta}{|F|}$

Taking a union bound over $f \in \mathcal{F}$, gives that with probability at least $1-\delta$,

$$\forall f \in \mathcal{F},\ \forall \tau \in [T], \quad \sum_{t=1}^{\tau} (f^\star(x^t,\pi^t) - r^t)^2 \le \sum_{t=1}^{\tau} (f(x^t,\pi^t) - r^t)^2 + \boxed{8\log(|\mathcal{F}|/\delta),} \tag{3.18}$$

and in particular

3.18 satisfies over $f \in F$, also $\hat{f}^\tau \in F$

$$\forall \tau \in [T+1], \quad \sum_{t=1}^{\tau-1} (f^\star(x^t,\pi^t) - r^t)^2 \le \sum_{t=1}^{\tau-1} (\boxed{\widehat{f}^\tau}(x^t,\pi^t) - r^t)^2 + 8\log(|\mathcal{F}|/\delta); \tag{3.19}$$

that is, we have $f^\star \in \mathcal{F}^\tau$ for all $\tau \in \{1, \dots, T+1\}$, proving the first claim. For the second part of the claim, observe that any $f \in \mathcal{F}^\tau$ must satisfy

$$\sum_{t=1}^{\tau-1} U^t(f) \le \beta$$

here $\beta \ge 8\log(|F|/\delta)$

since the empirical risk of $f^\star$ is never better than the empirical risk of the minimizer $\widehat{f}^t$. Thus from (3.16), with probability at least $1-\delta$, for all $\tau \le T$,

$$\sum_{t=1}^{\tau-1} \mathbb{E}_{t-1} U^t(f) \le 2\beta + 16\log(\delta^{-1}). \le 2\beta + 2\beta = 4\beta \tag{3.20}$$

The second claim follows by taking union bound over $f \in \mathcal{F}^\tau \subseteq \mathcal{F}$, and by (3.12). $\square$

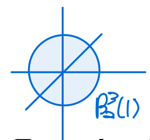
[10] We leave $\mathbb{E}_{t-1}$ on the right-hand side to include the case of randomized decisions $\pi^t \sim p^t$.

3.2 Optimism for Linear Models: The LinUCB Algorithm

We now instantiate the general template for optimistic algorithms developed in the previous section for the special case where $\mathcal{F}$ is a class of *linear functions*.

Linear models. We fix a *feature map* $\phi : \mathcal{X} \times \Pi \to \mathsf{B}_2^d(1)$, where $\mathsf{B}_2^d(1)$ is the unit-norm Euclidean ball in $\mathbb{R}^d$. The feature map is assumed to be known to the learning agent. For example, in the case of medical treatments, ϕ transforms the medical history and symptoms x for the patient, along with a possible treatment π, to a representation $\phi(x, \pi) \in \mathsf{B}_2^d(1)$. We take $\mathcal{F}$ to be the set of linear functions given by

$$\mathcal{F} = \{(x, \pi) \mapsto \langle \theta, \phi(x, \pi) \rangle \mid \theta \in \Theta\}, \tag{3.21}$$



where $\Theta \subseteq \mathsf{B}_2^d(1)$ is the parameter set. As before, we assume $f^\star \in \mathcal{F}$; we let θ^* denote the corresponding parameter vector, so that $f^\star(x, \pi) := \langle \theta^\star, \phi(x, \pi) \rangle$. With some abuse of notation, we associate the set of parameters Θ to the corresponding functions in $\mathcal{F}$.

To apply the technical results in the previous section, we assume for simplicity that $|\Theta| = |\mathcal{F}|$ is finite. To extend our results to potentially non-finite sets, one can work with an ε-discretization, or ε-net, which is of size at most $O(\varepsilon^{-d})$ using standard arguments. Taking $\varepsilon \sim 1/T$ ensures only a constant loss in cumulative regret relative to the continuous set of parameters, while $\log |\mathcal{F}| \lesssim d \log T$.

The LinUCB algorithm. The following figure displays an algorithm we refer to as LinUCB [12, 26, 1], which adapts the generic template for optimistic algorithms to the case where $\mathcal{F}$ is linear in the sense of (3.21).

LinUCB
Input: $\beta > 0$
for $t = 1, \ldots, T$ **do**
Compute the least squares solution $\widehat{\theta}^t$ (over $\theta \in \Theta$) given by

$$\widehat{\theta}^t = \arg\min_{\theta \in \Theta} \sum_{i<t} \left(\langle \theta, \phi(x^i, \pi^i) \rangle - r^i\right)^2.$$

Define

$$\widetilde{\Sigma}^t = \sum_{i=1}^{t-1} \phi(x^i, \pi^i)\phi(x^i, \pi^i)^\top + I.$$

Given x^t, select action

$$\pi^t \in \arg\max_{\pi \in \Pi} \max_{\theta : \|\theta - \widehat{\theta}^t\|^2_{\widetilde{\Sigma}^t} \leq 16\beta + 4} \langle \theta, \phi(x^t, \pi) \rangle .$$

Observe reward r^t

The following result shows that LinUCB enjoys a regret bound that scales with the complexity $\log|\mathcal{F}|$ of the model class and the feature dimension d.

Proposition 7: Let $\Theta \subseteq \mathsf{B}_2^d(1)$ and fix $\phi : \mathcal{X} \times \Pi \to \mathsf{B}_2^d(1)$. For a finite set $\mathcal{F}$ of linear functions (3.21), taking $\beta = 8\log(|\mathcal{F}|/\delta)$, LinUCB satisfies, with probability at least $1-\delta$,

$$\mathbf{Reg} \lesssim \sqrt{\beta d T \log(1+T/d)} \lesssim \sqrt{dT \log(|\mathcal{F}|/\delta) \log(1+T/d)}$$

for any sequence of contexts $x^1, \ldots, x^T$. More generally, for infinite $\mathcal{F}$, we may take $\beta = O(d\log(T))$[a] and

$$\mathbf{Reg} \lesssim d\sqrt{T}\log(T).$$

[a]This follows from a simple covering number argument.

Notably, this regret bound has no explicit dependence on the context space size $|\mathcal{X}|$. Interestingly, the bound is also independent of the number of actions $|\Pi|$, which is replaced by the dimension d; this reflects that the linear structure of $\mathcal{F}$ allows the learner to generalize not just across contexts, but across decisions. We will expand upon the idea of generalizing across actions in Section 4.

Proof of Proposition 7. The confidence set (3.9) in the generic optimistic algorithm template is

$$\mathcal{F}^t = \left\{ \theta \in \Theta : \sum_{i=1}^{t-1} (\underbrace{\langle \theta, \phi(x^i, \pi^i) \rangle}_{f} - r^i)^2 \le \sum_{i=1}^{t-1} (\langle \widehat{\theta}^t, \phi(x^i, \pi^i) \rangle - r^i)^2 + \beta \right\}, \tag{3.22}$$

where $\widehat{\theta}^t$ is the least squares solution computed in LinUCB. According to Lemma 10, with probability at least $1-\delta$, for all $t \in [T]$, all $\theta \in \mathcal{F}^t$ satisfy

$\sum (f(x^i,\pi^i) - f^*(x^i,\pi^i))^2 \le 4B$

by linear properties of $f \in$ LinUCB

$f - f^* = \langle \theta, \phi(x^i,\pi^i) \rangle - \langle \theta^*, \phi(x^i,\pi^i) \rangle$

$= \langle \theta - \theta^*, \phi(x^i,\pi^i) \rangle$

$$\sum_{i=1}^{t-1} (\langle \theta - \theta^*, \phi(x^i, \pi^i) \rangle)^2 \le 4\beta, \tag{3.23}$$

which means that $\mathcal{F}^t$ is a subset of[11]

square sum.

$$F^t \subseteq \Theta' = \left\{ \theta \in \Theta : \|\theta - \theta^*\|_{\Sigma^t}^2 \le 4\beta \right\}, \quad \text{where} \quad \Sigma^t = \sum_{i=1}^{t-1} \phi(x^i, \pi^i)\phi(x^i, \pi^i)^\top. \tag{3.24}$$

$(\theta - \theta^*)^T \Sigma (\theta - \theta^*) \le 16\beta$ ①

Since $\widehat{\theta}^t \in \mathcal{F}^t$, we have that for any $\theta \in \Theta'$, by triangle inequality, $\left\|\theta - \widehat{\theta}^t\right\|_{\Sigma^t}^2 \le 16\beta$. Furthermore, since $\widehat{\theta}^t \in \Theta \subseteq \mathsf{B}_2^d(1)$, $\left\|\theta - \widehat{\theta}^t\right\|_2 \le 2$. Combining the two constraints into one, we find that Θ' is a subset of

$(\theta - \hat{\theta}^t)^T I (\theta - \hat{\theta}^t) \le 4$ ②

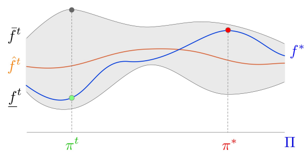
$$\Theta' \subseteq \Theta'' = \left\{ \theta \in \mathbb{R}^d : \left\|\theta - \widehat{\theta}^t\right\|_{\widetilde{\Sigma}^t}^2 \le 16\beta + 4 \right\}, \quad \text{where} \quad \widetilde{\Sigma}^t = \sum_{i=1}^{t-1} \phi(x^i, \pi^i)\phi(x^i, \pi^i)^\top + I. \tag{3.25}$$

① + ② $\Rightarrow (\theta - \hat{\theta}^t)^T (\Sigma + I)(\theta - \hat{\theta}^t) \le 16\beta + 4.$

The definition of $\bar{f}^t$ in (3.4) and the inclusion $\Theta' \subseteq \Theta''$ implies that

$$\bar{f}^t(x, \pi) \le \max_{\theta : \left\|\theta - \widehat{\theta}^t\right\|_{\widetilde{\Sigma}^t} \le \sqrt{16\beta + 4}} \langle \theta, \phi(x, \pi) \rangle = \left\langle \widehat{\theta}^t, \phi(x, \pi) \right\rangle + \sqrt{16\beta + 4}\, \|\phi(x, \pi)\|_{(\widetilde{\Sigma}^t)^{-1}}, \tag{3.26}$$

[11]For a PSD matrix $\Sigma \succeq 0$, we define $\|x\|_\Sigma = \sqrt{\langle x, \Sigma x \rangle}$.



and similarly $\underline{f}^t(x,\pi) \geq \langle \widehat{\theta}^t, \phi(x,\pi)\rangle - \sqrt{16\beta+4}\|\phi(x,\pi)\|_{(\widetilde{\Sigma}^t)^{-1}}$. We conclude that regret of the UCB algorithm, in view of Lemma 7, is

$$\mathbf{Reg} \lesssim 2\sqrt{16\beta+4}\sum_{t=1}^{T}\|\phi(x^t,\pi^t)\|_{(\widetilde{\Sigma}^t)^{-1}} \lesssim \sqrt{\beta T\sum_{t=1}^{T}\|\phi(x^t,\pi^t)\|^2_{(\widetilde{\Sigma}^t)^{-1}}}. \tag{3.27}$$

$\lesssim d\sqrt{T}\log(T)$

The above upper bound has the same flavor as the one in Lemma 8: as we obtain more and more information in some direction v, the matrix $\widetilde{\Sigma}^t$ has a larger and larger component in that direction, and for that direction v, the term $\|v\|^2_{(\widetilde{\Sigma}^t)^{-1}}$ becomes smaller and smaller. To conclude, we apply a potential argument, Lemma 11 below, to bound

$$\sum_{t=1}^{T}\|\phi(x^t,\pi^t)\|^2_{(\widetilde{\Sigma}^t)^{-1}} \lesssim d\log(1+T/d).$$

□

The following result is referred to as the elliptic potential lemma, and it can be thought of as a generalization of Lemma 8.

Lemma 11 (Elliptic potential lemma): Let $a_1,\dots,a_T \in \mathbb{R}^d$ satisfy $\|a_t\| \leq 1$ for all $t\in[T]$, and let $V_t = I + \sum_{s\leq t} a_s a_s^\top$. Then

$$\sum_{t=1}^{T}\|a_t\|^2_{V_{t-1}^{-1}} \leq 2d\log(1+T/d). \tag{3.28}$$

Proof Lemma 11 (sketch). First, the determinant of V_t evolves as

$$\det(V_t) = \det(V_{t-1})\left(1+\|a_t\|^2_{V_{t-1}^{-1}}\right).$$

Second, using the identity $u \wedge 1 \leq 2\ln(1+u)$ for $u\geq 0$, the left-hand side of (3.28) is at most $2\sum_{t=1}^{T}\log\left(1+\|a_t\|^2_{V_{t-1}^{-1}}\right)$. The proof concludes by upper bounding the determinant of V_n via the AM-GM inequality. We leave the details as an exercise; see also Lattimore and Szepesvári [60]. □

3.3 Moving Beyond Linear Classes: Challenges

We now present an example of a class $\mathcal{F}$ for which optimistic methods necessarily incur regret that scales linearly with either the cardinality of $\mathcal{F}$ or with cardinality of $\mathcal{X}$, meaning that we do not achieve the desired $\log|\mathcal{F}|$ scaling of regret that one might expect in (offline or online) supervised learning.

Example 3.1 (Failure of optimism for contextual bandits [37]). Let $A = 2$, and let $N \in \mathbb{N}$ be given. Let π_g and π_b be two actions available in each context, so that $\Pi = \{\pi_g, \pi_b\}$. Let $\mathcal{X} = \{x_1,\dots,x_N\}$ be a set of distinct contexts, and define a class $\mathcal{F} = \{f^\star, f_1,\dots,f_N\}$ of cardinality $N+1$ as follows. Fix $0<\varepsilon<1$. Let $f^\star(x,\pi_g) = 1-\varepsilon$ and $f^\star(x,\pi_b) = 0$ for any $x\in\mathcal{X}$. For each $i\in[N]$, $f_i(x_j,\pi_g) = 1-\varepsilon$ and $f_i(x_j,\pi_b) = 0$ for $j\neq i$, while $f_i(x_i,\pi_g) = 0$ and $f_i(x_i,\pi_b) = 1$.

①
②

Now, consider a (well-specified) problem instances in which rewards are deterministic and given by

$$r^t = f^\star(x^t, \pi^t),$$

which we note is a constant function with respect to the context. Since $f^\star$ is the true model, π_g is always the best action, bringing a reward of $1-\varepsilon$ per round. Any time π_b is chosen, the decision-maker incurs instantaneous regret $1-\varepsilon$. We will now argue that if we apply the generic optimistic algorithm from Section 3.1, it will choose π_b every time a new context is encountered, leading to $\Omega(N)$ regret.

bad action

Let S^t be the set of distinct contexts encountered before round t. Clearly, the exact minimizers of empirical square loss (see (3.8)) are $f^\star$, and all f_i where i is such that $x_i \notin S^t$. Hence, for any choice of $\beta \geq 0$, the confidence set in (3.9) contains all f_i for which $x_i \notin S^t$. This implies that for each $t \in [T]$ where $x^t = x_i \notin S^t$, action π_b has a higher upper confidence bound than π_g, since

$$\bar{f}^t(x^t, \pi_b) = f_i(x_i, \pi_b) = 1 > \bar{f}^t(x^t, \pi_g) = f^\star(x^t, \pi_g) = 1-\varepsilon.$$

Hence, the cumulative regret grows by $1-\varepsilon$ every time a new context is presented, and thus scales as $\Omega(N(1-\varepsilon))$ if the contexts are presented in order. That is, since $N = |\mathcal{X}| = |\mathcal{F}|-1$, the confidence-based algorithm fails to achieve logarithmic dependence on $\mathcal{F}$ (note that we may take $\varepsilon = 1/2$ for concreteness).

Let us remark that this failure continues even if contexts are stochastic. If the contexts are chosen via the uniform distribution on $\mathcal{X}$, then for $T \geq N$, at least a constant proportion of the domain will be presented, which still leads to a lower bound of

$$\mathbb{E}[\mathbf{Reg}] = \Omega(N) = \Omega(\min\{|\mathcal{F}|, |\mathcal{X}|\}).$$

◁

What is behind the failure of optimism in this example? The structure of $\mathcal{F}$ forces optimistic methods to *over-explore*, as the algorithm puts too much hope into trying the arm π_b for each new context. As a result, the confidence widths in (3.7) do not shrink quickly enough. Below, we will see that there are alternative methods which *do* enjoy logarithmic dependence on the size of $\mathcal{F}$, with the best of these methods achieving regret $O(\sqrt{AT\log|\mathcal{F}|})$.

We mention in passing that even though optimism does not succeed in general, it is useful to understand in what cases it works. We saw that the structure of linear classes in $\mathbb{R}^d$ only allowed for d "different" directions, while in the example above, the optimistic algorithm gets tricked by each new context, and is not able to shrink the confidence band quickly enough over the domain. In a few lectures (Section 4), we will introduce the eluder dimension, a structural property of the class $\mathcal{F}$ which is sufficient for optimistic methods to experience low regret, generalizing the positive result for the linear setting.

3.4 The ε-Greedy Algorithm for Contextual Bandits

Given that the principle of optimism only leads to low regret for classes $\mathcal{F}$ with special structure, we are left wondering whether there are more general algorithmic principles for decision making that can succeed for *any* class $\mathcal{F}$. In this section and the following one, we will present two such principles. Both approaches will still make use of supervised learning with the class $\mathcal{F}$, but will build upon *online* supervised learning as opposed to offline/statistical learning. To make the use of supervised learning as modular as possible, we will abstract this away using the notion of an *online regression oracle* [36].

Definition 3 (Online Regression Oracle): At each time $t \in [T]$, an *online regression oracle* returns, given 1~t-1.

$$(x^1, \pi^1, r^1), \ldots, (x^{t-1}, \pi^{t-1}, r^{t-1})$$

with $\mathbb{E}[r^i | x^i, \pi^i] = f^\star(x^i, \pi^i)$ and $\pi^i \sim p^i$, a function $\widehat{f}^t : \mathcal{X} \times \Pi \to \mathbb{R}$ such that

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t} (\widehat{f}^t(x^t, \pi^t) - f^\star(x^t, \pi^t))^2 \le \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)$$

with probability at least $1 - \delta$. For the results that follow, $p^i = p^i(\cdot | x^i, \mathcal{H}^{i-1})$ will represent the randomization distribution of a decision-maker.

과거 가설공간.

For example, for finite classes, the (averaged) exponential weights method introduced in Section 1.6 is an online regression oracle with $\boxed{\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta) = \log(|\mathcal{F}|/\delta).}$ More generally, in view of Lemma 6, any online learning algorithm that attains low square loss regret for the problem of predicting of r^t based on (x^t, π^t) leads to a valid online regression oracle.

Note that we make use of *online learning* oracles for the results that follow because we aim to derive regret bounds that hold for arbitrary, potentially adversarial sequences $x^1, \ldots, x^T$. If we instead assume that contexts are i.i.d., it is reasonable to make use of algorithms for "*offline estimation*", or "statistical learning with $\mathcal{F}$." See Section 3.5.1 for further discussion.

The first general-purpose contextual bandit algorithm we will study, illustrated below, is a contextual counterpart to the ε-Greedy method introduced in Section 2.

ε-Greedy for Contextual Bandits

Input: $\varepsilon \in (0, 1)$.
for $t = 1, \ldots, T$ **do**
  Obtain $\widehat{f}^t$ from online regression oracle for $(x^1, \pi^1, r^1), \ldots, (x^{t-1}, \pi^{t-1}, r^{t-1})$.
  Observe x^t.
  With prob. ε, select $\pi^t \sim \mathrm{unif}([A])$, and with prob. $1 - \varepsilon$, choose the greedy action

$$\widehat{\pi}^t = \arg\max_{\pi \in [A]} \widehat{f}^t(x^t, \pi).$$

  Observe reward r^t.

At each step t, the algorithm uses an online regression oracle to compute a reward estimator $\widehat{f}^t(x, a)$ based on the data $\mathcal{H}^{t-1}$ collected so far. Given this estimator, the algorithm uses the same sampling strategy as in the non-contextual case: with probability $1 - \varepsilon$, the algorithm chooses the greedy decision

$$\boxed{\widehat{\pi}^t} = \arg\max_{\pi} \widehat{f}^t(x^t, \pi), \tag{3.29}$$

and with probability ε it samples a uniform random action $\pi^t \sim \mathrm{unif}(\{1, \ldots, A\})$. The following theorem shows that whenever the online estimation oracle has low estimation error $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)$, this method achieves low regret.

Proposition 8: Assume $f^\star \in \mathcal{F}$ and $f^\star(x,a) \in [0,1]$. Suppose the decision-maker has access to an online regression oracle (Definition 3) with a guarantee $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F},T,\delta)$. Then by choosing ε appropriately, the ε-Greedy algorithm ensures that with probability at least $1-\delta$,

$$\mathbf{Reg} \lesssim A^{1/3}T^{2/3} \cdot \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F},T,\delta)^{1/3}$$

for any sequence $x^1,\dots,x^T$. As a special case, when $\mathcal{F}$ is finite, if we use the (averaged) exponential weights algorithm as an online regression oracle, the ε-Greedy algorithm has

$$\mathbf{Reg} \lesssim A^{1/3}T^{2/3} \cdot \log^{1/3}(|\mathcal{F}|/\delta).$$

Notably, this result scales with $\log|\mathcal{F}|$ for *any finite class*, analogous to regret bounds for offline/online supervised learning. The $T^{2/3}$-dependence in the regret bound is suboptimal (as seen for the special case of non-contextual bandits), which we will address using more deliberate exploration methods in the sequel.

Proof of Proposition 8. Recall that p^t denotes the randomization strategy on round t, computed after observing x^t. Following the same steps as the proof of Proposition 4, we can bound regret by

$$\mathbf{Reg} = \sum_{t=1}^{T} \mathbb{E}_{\pi^t\sim p^t}[f^\star(x^t,\pi^\star(x^t)) - f^\star(x^t,\pi^t)] \le \sum_{t=1}^{T} f^\star(x^t,\pi^\star(x^t)) - f^\star(x^t,\widehat{\pi}^t) + \varepsilon T,$$

ε-greedy.

where the εT term represents the bias incurred by exploring uniformly.

Fix t and abbreviate $\pi^\star(x^t) = \pi^\star$. We have

$$\begin{aligned}
&f^\star(x^t,\pi^\star) - f^\star(x^t,\widehat{\pi}^t)\\
&= [f^\star(x^t,\pi^\star) - \widehat{f}^t(x^t,\pi^\star)] + [\widehat{f}^t(x^t,\pi^\star) - \widehat{f}^t(x^t,\widehat{\pi}^t)] + [\widehat{f}^t(x^t,\widehat{\pi}^t) - f^\star(x^t,\widehat{\pi}^t)]\\
&\le \sum_{\pi\in\{\widehat{\pi}^t,\pi^\star\}} |f^\star(x^t,\pi) - \widehat{f}^t(x^t,\pi)|\\
&= \sum_{\pi\in\{\widehat{\pi}^t,\pi^\star\}} \frac{1}{\sqrt{p^t(\pi)}}\sqrt{p^t(\pi)}|f^\star(x^t,\pi) - \widehat{f}^t(x^t,\pi)|.
\end{aligned}$$

By the Cauchy-Schwarz inequality, the last expression is at most

$$\left\{\sum_{\pi\in\{\widehat{\pi}^t,\pi^\star\}} \frac{1}{p^t(\pi)}\right\}^{1/2} \left\{\sum_{\pi\in\{\widehat{\pi}^t,\pi^\star\}} p^t(\pi)\Big(f^\star(x^t,\pi) - \widehat{f}^t(x^t,\pi)\Big)^2\right\}^{1/2} \tag{3.30}$$

$$\le \sqrt{\frac{2A}{\varepsilon}}\left\{\mathbb{E}_{\pi^t\sim p^t}\Big(f^\star(x^t,\pi^t) - \widehat{f}^t(x^t,\pi^t)\Big)^2\right\}^{1/2}. \tag{3.31}$$

Since $p^t(\pi) \ge \frac{\varepsilon}{A}$

$\sum \frac{1}{p^t(\pi)} \le \frac{A}{\varepsilon} + \frac{A}{\varepsilon} = \frac{2A}{\varepsilon}$

1~T

Summing across t, this gives

$$\sum_{t=1}^{T} f^\star(x^t, \pi^\star(x^t)) - f^\star(x^t, \widehat{\pi}^t) \le \sqrt{\frac{2A}{\varepsilon}} \sum_{t=1}^{T} \left\{ \mathbb{E}_{\pi^t \sim p^t} \left(f^\star(x^t, \pi^t) - \widehat{f}^t(x^t, \pi^t) \right)^2 \right\}^{1/2} \tag{3.32}$$

$$\le \sqrt{\frac{2AT}{\varepsilon}} \left\{ \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t} \left(f^\star(x^t, \pi^t) - \widehat{f}^t(x^t, \pi^t) \right)^2 \right\}^{1/2}. \tag{3.33}$$

where $T > 0$

Now observe that the online regression oracle guarantees that with probability $1 - \delta$,

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t} \left(f^\star(x^t, \pi^t) - \widehat{f}^t(x^t, \pi^t) \right)^2 \le \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta).$$

Whenever this occurs, we have

$$\mathbf{Reg} \lesssim \sqrt{\frac{AT\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)}{\varepsilon}} + \varepsilon T.$$

Choosing ε to balance the two terms leads to the claimed result. □

3.5 Inverse Gap Weighting: An Optimal Algorithm for General Model Classes

To conclude this section, we present a general, oracle-based algorithm for contextual bandits which achieves

$$\mathbf{Reg} \lesssim \sqrt{AT \log|\mathcal{F}|}$$

for any finite class $\mathcal{F}$. As with ε-Greedy, this approach has no dependence on the cardinality $|\mathcal{X}|$ of the context space, reflecting the ability to generalize across contexts. The dependence on T improves upon ε-Greedy, and is optimal.

To motivate the approach, recall that conceptually, the key step of the proof of Proposition 8 involved relating the instantaneous regret

$$\mathbb{E}_{\pi^t \sim p^t}[f^\star(x^t, \pi^\star(x^t)) - f^\star(x^t, \pi^t)] \tag{3.34}$$

of the decision maker at time t to the instantaneous estimation error

$$\mathbb{E}_{\pi^t \sim p^t}\left[\left(f^\star(x^t, \pi^t) - \widehat{f}^t(x^t, \pi^t) \right)^2 \right] \tag{3.35}$$

between $\widehat{f}^t$ and $f^\star$ under the randomization distribution p^t. The ε-Greedy exploration distribution gives a way to relate these quantities, but the algorithm's regret is suboptimal because the randomization distribution puts mass at least ε/A on every action, even those that are clearly suboptimal and should be discarded. One can ask whether there exists a better randomization strategy that still admits an upper bound on (3.34) in terms of (3.35). Proposition 9 below establishes exactly that. At first glance, this distribution might appear to be somewhat arbitrary or "magical", but we will show in subsequent chapters that it arises as a special case of more general—and in some sense, universal—principle for designing decision making algorithms, which extends well beyond contextual bandits.

Definition 4 (Inverse Gap Weighting [2, 36]): Given a vector $\widehat{f} = (\widehat{f}(1), \dots, \widehat{f}(A)) \in \mathbb{R}^A$, the Inverse Gap Weighting distribution $p = \mathsf{IGW}_\gamma(\widehat{f}(1), \dots, \widehat{f}(A))$ with parameter $\gamma \geq 0$ is defined as

$$p(\pi) = \frac{1}{\lambda + 2\gamma(\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi))}, \tag{3.36}$$

where $\widehat{\pi} = \arg\max_\pi \widehat{f}(\pi)$ is the greedy action, and where $\lambda \in [1, A]$ is chosen such that $\sum_\pi p(\pi) = 1$.

Above, the normalizing constant $\lambda \in [1, A]$ is always guaranteed to exist, because we have $\frac{1}{\lambda} \leq \sum_\pi p(\pi) \leq \frac{A}{\lambda}$, and because $\lambda \mapsto \sum_\pi p(\pi)$ is continuous over $[1, A]$.

Let us give some intuition behind the distribution in (3.36). We can interpret the parameter γ as trading off exploration and exploitation. Indeed, $\gamma \to 0$ gives a uniform distribution, while $\gamma \to \infty$ amplifies the gap between the greedy action $\widehat{\pi}$ and any action with $\widehat{f}(\pi) < \widehat{f}(\widehat{\pi})$, resulting in a distribution supported only on actions that achieve the largest estimated value $\widehat{f}(\widehat{\pi})$.

- $\gamma \to 0$, $p \to \frac{1}{\lambda}$
- $\gamma \to \infty$, $p \to 0$

The following fundamental technical result shows that playing the Inverse Gap Weighting distribution always suffices to link the instantaneous regret in (3.34) in to the instantaneous estimation error in (3.35).

Proposition 9: Consider a finite decision space $\Pi = \{1, \dots, A\}$. For any vector $\widehat{f} \in \mathbb{R}^A$ and $\gamma > 0$, define $\boxed{p = \mathsf{IGW}_\gamma(\widehat{f}(1), \dots, \widehat{f}(A))}$. This strategy guarantees that for all $f^\star \in \mathbb{R}^A$,

$$\mathbb{E}_{\pi\sim p}[f^\star(\pi^\star) - f^\star(\pi)] \leq \frac{A}{\gamma} + \gamma \cdot \mathbb{E}_{\pi\sim p}\left[(\widehat{f}(\pi) - f^\star(\pi))^2\right]. \tag{3.37}$$

Proof of Proposition 9. We break the "regret" term on the left-hand side of (3.37) into three terms:

$$\left(\frac{f^\star(x^t,\pi^\star) - f^\star(x^t,\widehat{\pi}^t)}{= [f^\star(x^t,\pi^\star) - \widehat{f}^t(x^t,\pi^\star)] + [\widehat{f}^t(x^t,\pi^\star) - \widehat{f}^t(x^t,\widehat{\pi}^t)] + [\widehat{f}^t(x^t,\widehat{\pi}^t) - f^\star(x^t,\widehat{\pi}^t)]} \right)$$

$$\mathbb{E}_{\pi\sim p}\big[f^\star(\pi^\star) - f^\star(\pi)\big] = \underbrace{\mathbb{E}_{\pi\sim p}\big[\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi)\big]}_{\text{(I) exploration bias}} + \underbrace{\mathbb{E}_{\pi\sim p}\big[\widehat{f}(\pi) - f^\star(\pi)\big]}_{\text{(II) est. error on policy}} + \underbrace{f^\star(\pi^\star) - \widehat{f}(\widehat{\pi})}_{\text{(III) est. error at opt}}.$$

The first term asks "how much would we lose by exploring, if $\widehat{f}$ were the true reward function?", and is equal to

$$\boxed{\sum_\pi \frac{\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi)}{\lambda + 2\gamma\big(\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi)\big)} \leq \frac{A-1}{2\gamma},}$$

while the second term is at most

$$\boxed{\sqrt{\mathbb{E}_{\pi\sim p}\big[(\widehat{f}(\pi) - f^\star(\pi))^2\big]} \leq \frac{1}{2\gamma} + \frac{\gamma}{2}\mathbb{E}_{\pi\sim p}(\widehat{f}(\pi) - f^\star(\pi))^2.}$$

The third term can be further written as

$$\begin{aligned}
\boxed{f^\star(\pi^\star) - \widehat{f}(\pi^\star) - (\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star))} &\leq \frac{\gamma}{2}p(\pi^\star)(f^\star(\pi^\star) - \widehat{f}(\pi^\star))^2 + \frac{1}{2\gamma p(\pi^\star)} - (\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star)) \\
&\leq \frac{\gamma}{2}\mathbb{E}_{\pi\sim p}(f^\star(\pi) - \widehat{f}(\pi))^2 + \left[\frac{1}{2\gamma p(\pi^\star)} - (\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star))\right].
\end{aligned}$$

The term in brackets above is equal to

$\lambda \in [1, A]$

$$\frac{\lambda + 2\gamma(\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star))}{2\gamma} - (\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star)) = \frac{\lambda}{2\gamma} \leq \frac{A}{2\gamma}.$$

use $p(\pi^\star) = \lambda + 2\gamma(\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star))$

□

The simple result we just proved is remarkable. The special IGW strategy guarantees a relation between regret and estimation error for *any* estimator $\widehat{f}$ and any $f^\star$, irrespective of the problem structure or the class $\mathcal{F}$. Proposition 9 will be at the core of the development for the rest of the course, and will be greatly generalized to general decision making problems and reinforcement learning.

Below, we present a contextual bandit algorithm called SquareCB [36] which makes use of the Inverse Gap Weighting distribution.

SquareCB

Input: Exploration parameter $\gamma > 0$.

for $t = 1, \dots, T$ **do**

Obtain $\widehat{f}^t$ from online regression oracle with $(x^1, \pi^1, r^1), \dots, (x^{t-1}, \pi^{t-1}, r^{t-1})$.

Observe x^t.

Compute $p^t = \mathsf{IGW}_\gamma\left(\widehat{f}^t(x^t, 1), \dots, \widehat{f}^t(x^t, A)\right)$. $= \frac{1}{\lambda + 2\gamma(\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star))}$

Select action $\pi^t \sim p^t$.

Observe reward r^t.

At each step t, the algorithm uses an online regression oracle to compute a reward estimator $\widehat{f}^t(x, a)$ based on the data $\mathcal{H}^{t-1}$ collected so far. Given this estimator, the algorithm uses Inverse Gap Weighting to compute $p^t = \mathsf{IGW}_\gamma(\widehat{f}^t(x^t, \cdot))$ as an exploratory distribution, then samples $\pi^t \sim p^t$.

The following result, which is a near-immediate consequence of Proposition 9, gives a regret bound for this algorithm.

Proposition 10: Given a class $\mathcal{F}$ with $f^\star \in \mathcal{F}$, assume the decision-maker has access to an online regression oracle (Definition 3) with estimation error $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)$. Then SquareCB with $\gamma = \sqrt{TA/\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)}$ attains a regret bound of

$$\mathbf{Reg} \lesssim \sqrt{A T \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)}$$

with probability at least $1 - \delta$ for any sequence $x^1, \dots, x^T$. As a special case, when $\mathcal{F}$ is finite, the averaged exponential weights algorithm achieves $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta) \lesssim \log(|\mathcal{F}|/\delta)$, leading to

$$\mathbf{Reg} \lesssim \sqrt{AT \log(|\mathcal{F}|/\delta)}.$$

Proof of Proposition 10. We begin with regret, then add and subtract the squared estimation error as follows:

$$\mathbf{Reg} = \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(x^t, \pi^\star) - f^\star(x^t, \pi^t)]$$

$*\mathrm{Est}_{Sq}(\mathcal{F}, T, \delta) = \log(|\mathcal{F}|/\delta)$

$$= \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[f^\star(x^t, \pi^\star) - f^\star(x^t, \pi^t) - \gamma \cdot (f^\star(x^t, \pi^t) - \widehat{f}^t(x^t, \pi^t))^2\right] + \gamma \cdot \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta).$$

$\left(\mathbb{E}_{\pi\sim p}[f^\star(\pi^\star) - f^\star(\pi)] \leq \frac{A}{\gamma} + \gamma \cdot \mathbb{E}_{\pi\sim p}[(\widehat{f}(\pi) - f^\star(\pi))^2].\right)$

By appealing to Proposition 9 with $\widehat{f}(x^t,\cdot)$ and $f^\star(x^t,\cdot)$, for each step t, we have

$$\mathbb{E}_{\pi^t\sim p^t}\left[f^\star(x^t,\pi^\star) - f^\star(x^t,\pi^t) - \gamma\cdot(f^\star(x^t,\pi^t) - \widehat{f}^t(x^t,\pi^t))^2\right] \leq \frac{A}{\gamma},$$

and thus

$$\mathbf{Reg} \leq \frac{TA}{\gamma} + \gamma\cdot\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F},T,\delta).$$

Choosing γ to balance these terms yields the result. $\square$

If the online regression oracle is minimax optimal (that is, $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F},T,\delta)$ is the "best possible" for $\mathcal{F}$) then SquareCB is also minimax optimal for $\mathcal{F}$. Thus, IGW not only provides a connection between online supervised learning and decision making, but it does so in an optimal fashion. Establishing minimax optimality is beyond the scope of this course: it requires understanding of minimax optimality of online regression with arbitrary $\mathcal{F}$, as well as lower bound on regret of contextual bandits with arbitrary sequences of contexts. We refer to Foster and Rakhlin [36] for details.

3.5.1 Extending to Offline Regression

When $x^1,\ldots,x^T$ are i.i.d., it is natural to ask whether an online regression method that works for arbitrary sequences is necessary, or whether one can work with a weaker oracle tuned to i.i.d. data. For SquareCB, it turns out that any oracle for *offline regression* (defined below) is sufficient.

Definition 5 (Offline Regression Oracle): Given

$$(x^1,\pi^1,r^1),\ldots,(x^{t-1},\pi^{t-1},r^{t-1})$$

where $x^1,\ldots,x^{t-1}$ are i.i.d., $\pi^i\sim p(x^i)$ for fixed $p:\mathcal{X}\to\Delta(\Pi)$ and $\mathbb{E}[r^i|x^i,\pi^i] = f^\star(x^i,\pi^i)$, an *offline regression oracle* returns a function $\widehat{f}:\mathcal{X}\times\Pi\to\mathbb{R}$ such that

$$\mathbb{E}_{x,\pi\sim p(x)}(\widehat{f}(x,\pi) - f^\star(x,\pi))^2 \leq t^{-1}\mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F},t,\delta)$$

with probability at least $1-\delta$.

Note that the normalization t^{-1} above is introduced to keep the scaling consistent with our conventions for offline estimation.

Below, we state a variant of SquareCB which is adapted to offline oracles [79]. Compared to the SquareCB for online oracles, the main change is that we update the estimation oracle and exploratory distribution on an epoched schedule as opposed to updating at every round. In addition, the parameter γ for the Inverse Gap Weighting distribution changes as a function of the epoch.

SquareCB with offline oracles

Input: Exploration parameters $\gamma_1,\gamma_2,\ldots$ and epoch sizes $\tau_1,\tau_2,\ldots$

for $m=1,2,\ldots$ **do**

Obtain $\widehat{f}^m$ from offline regression oracle with

$(x^{\tau_{m-2}+1},\pi^{\tau_{m-2}+1},r^{\tau_{m-2}+1}),\ldots,(x^{\tau_{m-1}},\pi^{\tau_{m-1}},r^{\tau_{m-1}}).$

$\tau_{m+1} \sim \tau_m$.

for $t = \tau_{m-1}+1, \ldots, \tau_m$ **do**
  Observe x^t.
  Compute $p^t = \mathsf{IGW}_{\gamma_m}\left(\widehat{f}^m(x^t, 1), \ldots, \widehat{f}^m(x^t, A)\right)$.
  Select action $\pi^t \sim p^t$.
  Observe reward r^t.

While this algorithm is quite intuitive, proving a regret bound for it is quite non-trivial—much more so than the online oracle variant. They key challenge is that, while the contexts $x^1, \ldots, x^T$ are i.i.d., the decisions $\pi^1, \ldots, \pi^T$ evolve in a time-dependent fashion, which makes it unclear to invoke the guarantee in Definition 5. Nonetheless, the following remarkable result shows that this algorithm attains a regret bound similar to that of Proposition 10.

Proposition 11 (Simchi-Levi and Xu [79]): Let $\tau_m = 2^m$ and $\gamma_m = \sqrt{AT/\mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, \tau_{m-1}, \delta)}$ for $m = 1, 2, \ldots$. Then with probability at least $1-\delta$, regret of $\mathsf{SquareCB}$ with an offline oracle is at most

$$\mathbf{Reg} \lesssim \sum_{m=1}^{\lceil \log T \rceil} \sqrt{A \cdot \tau_m \cdot \mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, \tau_m, \delta/m^2)}.$$

Under mild assumptions, above bound scales as

$$\mathbf{Reg} \lesssim \sqrt{A \cdot T \cdot \mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, \tau_m, \delta/\log T)}.$$

For a finite class $\mathcal{F}$, we recall from Section 1 that empirical risk with the square loss (least squares) achieves $\mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, T, \delta) \lesssim \log(|\mathcal{F}|/\delta)$, which gives

$$\mathbf{Reg} \lesssim \sqrt{AT \log(|\mathcal{F}|/\delta)}.$$

3.6 Exercises

Exercise 5 (Unstructured Contextual Bandits): Consider a contextual bandit problem with a finite set $\mathcal{X}$ of possible contexts, and a finite set of actions $\mathcal{A}$. Show that running UCB independently for each context yields a regret bound of the order $\widetilde{O}(\sqrt{|X||\mathcal{A}|T})$ in expectation, ignoring logarithmic factors. In the setting where $\mathcal{F} = \mathcal{X} \times \mathcal{A} \to [0, 1]$ is unstructured, and consists of all possible functions, this is essentially optimal.

Exercise 6 (ε-Greedy with Offline Oracles): In Proposition 8, we analyzed the ε-Greedy contextual bandit algorithm assuming access to an online regression oracle. Because we appeal to online learning, this algorithm was able to handle adversarial contexts $x^1, \ldots, x^T$. In the present problem, we will modify the ε-greedy algorithm and proof to show that if contexts are stochastic (that is $x^t \sim \mathcal{D}\ \ \forall t$, where $\mathcal{D}$ is a fixed distribution), ε-greedy works even if we use an *offline oracle* (Definition 5).

We consider the following variant of ε-greedy. The algorithm proceeds in epochs $m = 0, 1, \ldots$ of doubling size

$$\{2\}, \{3, 4\}, \{5 \ldots 8\}, \ldots, \underbrace{\{2^m + 1, 2^{m+1}\}}_{\text{epoch } m}, \ldots, \{T/2 + 1, T\};$$

we assume without loss of generality that T is a power of 2, and that an arbitrary decision is made on round $t = 1$. At the end of each epoch $m-1$, the offline oracle is invoked with the data from the epoch, producing an estimated model $\widehat{f}^m$. This model is used for the greedy step in the next epoch m. In other words, for any round $t \in [2^m + 1, 2^{m+1}]$ of epoch m, the algorithm observes $x^t \sim \mathcal{D}$, chooses an action $\pi^t \sim \mathrm{unif}[A]$ with probability ε and chooses the greedy action

$$\pi^t = \arg\max_{\pi \in [A]} \widehat{f}^m(x^t, \pi)$$

with probability $1-\varepsilon$. Subsequently, the reward r^t is observed.

1. Prove that for any $T \in \mathbb{N}$ and $\delta > 0$, by setting ε appropriately, this method ensures that with probability at least $1-\delta$,

$$\mathbf{Reg} \lesssim A^{1/3} T^{1/3} \left(\sum_{m=1}^{\log_2 T} 2^{m/2} \mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, 2^{m-1}, \delta/m^2)^{1/2} \right)^{2/3}$$

2. Recall that for a finite class, ERM achieves $\mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, T, \delta) \lesssim \log(|\mathcal{F}|/\delta)$. Show that with this choice, the above upper bound matches that in Proposition 8, up to logarithmic in T factors.

Exercise 7 (Model Misspecification in Contextual Bandits): In Proposition 10, we showed that for contextual bandits with a general class $\mathcal{F}$, SquareCB attains regret

$$\mathbf{Reg} \lesssim \sqrt{AT \cdot \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)}. \tag{3.38}$$

To do so, we assumed that $f^\star \in \mathcal{F}$, where $f^\star(x,a) := \mathbb{E}_{r \sim M^\star(\cdot|x,a)}[r]$; that is, we have a well-specified model. In practice, it may be unreasonable to assume that we have $f^\star \in \mathcal{F}$. Instead, a weaker assumption is that there exists some function $\bar{f} \in \mathcal{F}$ such that

$$\max_{x \in \mathcal{X}, a \in \mathcal{A}} |\bar{f}(x,a) - f^\star(x,a)| \le \varepsilon$$

for some $\varepsilon > 0$; that is, the model is *ε-misspecified*. In this problem, we will generalize the regret bound for SquareCB to handle misspecification. Recall that in the lecture notes, we assumed (Definition 3) that the regression oracle satisfies

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - f^\star(x^t, \pi^t))^2 \right] \le \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta).$$

In the misspecified setting, this is too much to ask for. Instead, we will assume that the oracle satisfies the following guarantee for *every sequence*:

$$\sum_{t=1}^{T} (\widehat{f}^t(x^t, \pi^t) - r^t)^2 - \min_{f \in \mathcal{F}} \sum_{t=1}^{T} (f(x^t, \pi^t) - r^t)^2 \le \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T).$$

Whenever $f^\star \in \mathcal{F}$, we have $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta) \lesssim \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + \log(1/\delta)$ with probability at least $1-\delta$. However, it is possible to keep $\mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T)$ small even when $f^\star \notin \mathcal{F}$. For example, the averaged exponential weights algorithm satisfies this guarantee with $\mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) \lesssim \log|\mathcal{F}|$, regardless of whether $f^\star \in \mathcal{F}$.

We will show that for every $\delta > 0$, with an appropriate choice of γ, SquareCB (that is, the algorithm that chooses $p^t = \mathsf{IGW}_\gamma(\widehat{f}^t(x^t, \cdot)))$ ensures that with probability at least $1 - \delta$,

$$\mathbf{Reg} \lesssim \sqrt{AT \cdot (\mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + \log(1/\delta))} + \varepsilon \cdot A^{1/2} T.$$

Assume that all functions in $\mathcal{F}$ and rewards take values in $[0, 1]$.

1. Show that for any sequence of estimators $\widehat{f}^1, \ldots, \widehat{f}^t$, by choosing $p^t = \mathsf{IGW}_\gamma(\widehat{f}^t(x^t, \cdot))$, we have that

$$\mathbf{Reg} = \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(x^t, \pi^\star(x^t)) - f^\star(x^t, \pi^t)] \lesssim \frac{AT}{\gamma} + \gamma \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] + \varepsilon T.$$

If we had $f^\star = \bar{f}$, this would follow from Proposition 9, but the difference is that in general ($\bar{f} \neq f^\star$), the expression above measures estimation error with respect to the best-in-class model $\bar{f}$ rather than the true model $f^\star$ (at the cost of an extra εT factor).

2. Show that the following inequality holds for every sequence

$$\sum_{t=1}^{T}(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2 \leq \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + 2\sum_{t=1}^{T}(r^t - \bar{f}(x^t, \pi^t))(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t)).$$

3. Using Freedman's inequality (Lemma 35), show that with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] \leq 2\sum_{t=1}^{T}(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2 + O(\log(1/\delta)).$$

4. Using Freedman's inequality once more, show that with probability at least $1 - \delta$,

$$2\sum_{t=1}^{T}(r^t - \bar{f}(x^t, \pi^t))(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t)) \leq \frac{1}{4}\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] + O(\varepsilon^2 T + \log(1/\delta)).$$

Conclude that with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] \lesssim \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + \varepsilon^2 T + \log(1/\delta).$$

5. Combining the previous results, show that for any $\delta > 0$, by choosing $\gamma > 0$ appropriately, we have that with probability at least $1 - \delta$,

$$\mathbf{Reg} \lesssim \sqrt{AT \cdot (\mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + \log(1/\delta))} + \varepsilon \cdot A^{1/2} T.$$

4. STRUCTURED BANDITS

Up to this point, we have focused our attention on bandit problems (with or without contexts) in which the decision space Π is a small, finite set. This section introduces the *structured bandit* problem, which generalizes the basic (non-contextual) multi-armed bandit problem by allowing for large, potentially infinite or continuous decision spaces. The protocol for the setting is as follows.